Sequential Logic – Part 3
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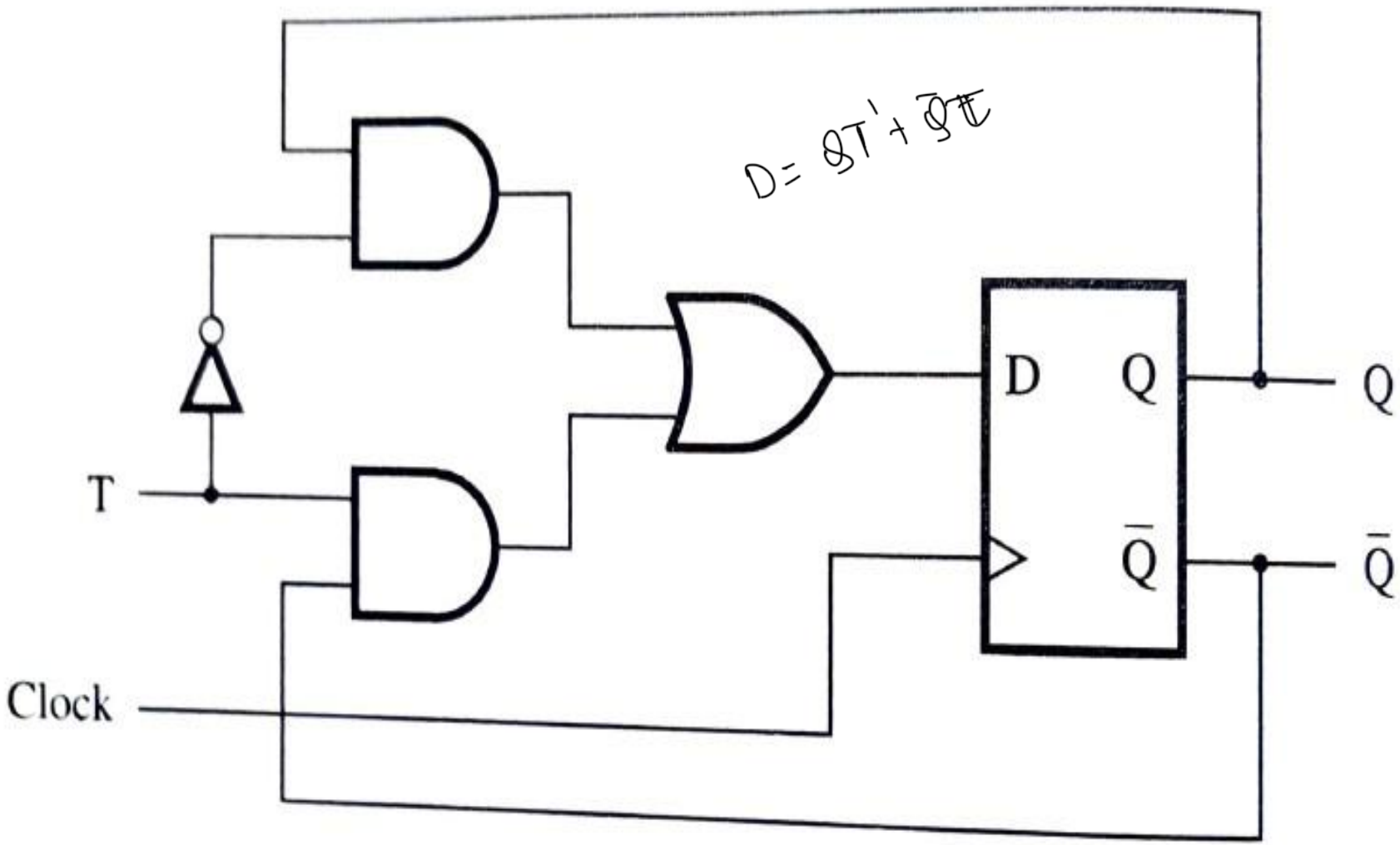
Review

- We have studied D flip-flops and two versions, namely master-slave D flip-flop and positive edge triggered D flip-flop.
- We will now look at a few more flip-flops and some aspects relating to reliable operation of the flip-flops.

Other flip-flops and their origins

- The D flip-flop is a versatile storage element and can be used to derive some other types of flip-flops.
- In particular, two flip-flops called T flip-flop and JK flip-flop can be derived from a positive edge triggered D flip-flop.

T flip-flop circuit



Explanation

- The feedback connections are such that input signal D is either equal to the value of Q or Qbar (Q') and this happens under the control of the signal labelled T.
- T stands for Toggle.
- If T = 0, D = Q and the state remains the same.
- However, if T = 1, then D = Qbar (or Q').

Some terms of interest

- For the basic flip-flops (namely D, T and a variation called JK to be discussed), there are two terms of interest: the characteristic table and characteristic equation. One can also describe the input via an equation.
- For the T flip-flop, the input equation is given by:

$$D = T \oplus Q = TQ' + T'Q$$

Can see the truth of this from the figure (look at the gates and observe that TQ' + T'Q is what we give to D)

The characteristic equation for T flip-flop

- It is given by

$$Q(t+1) = T \oplus Q = TQ' + T'Q$$

This equation says what the "next state" is in terms of T and the "present state". The characteristic equation will be of interest when we discuss state machines.

The characteristic table for T flip-flop

T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

By the way, for the D flip-flop we have:

D	Q(t + 1)	
0	0	Reset
1	1	Set

Characteristic Table

$$Q(t+1) = D$$

Characteristic equation

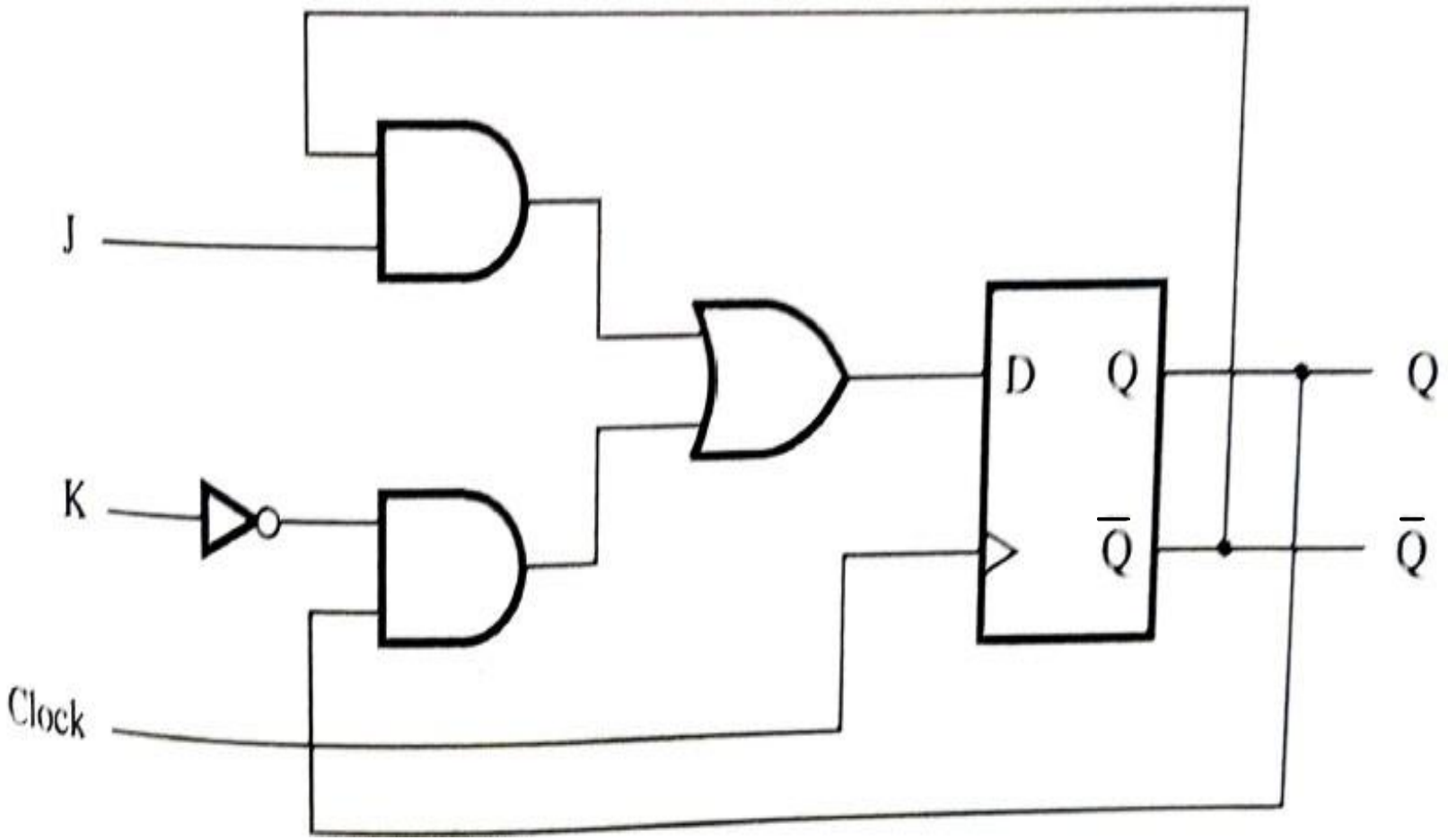
On the characteristic equation for a D flip-flop

- The characteristic equation of a D flip-flop shows a direct relationship between the D input and the next state.
- In other words, whatever is at D will appear at the Q output in the next clock cycle.
- This equation also allows us to examine the opposite direction. That is, consider the following question: “Suppose we knew the next state, then can we say what input condition caused it ? ”

Moving on ..

- We also have something called as the JK flip-flop.
- It is a variation of the T flip-flop.
- Instead of using a single control input T, we use two inputs, namely J and K.
- The circuit is shown next.

JK flip-flop circuit



JK f/f characteristic table

J	K	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$\overline{Q}(t)$

JK flip-flop equations

$$D = JQ' + K'Q$$

Input equation

$$Q(t+1) = JQ' + K'Q$$

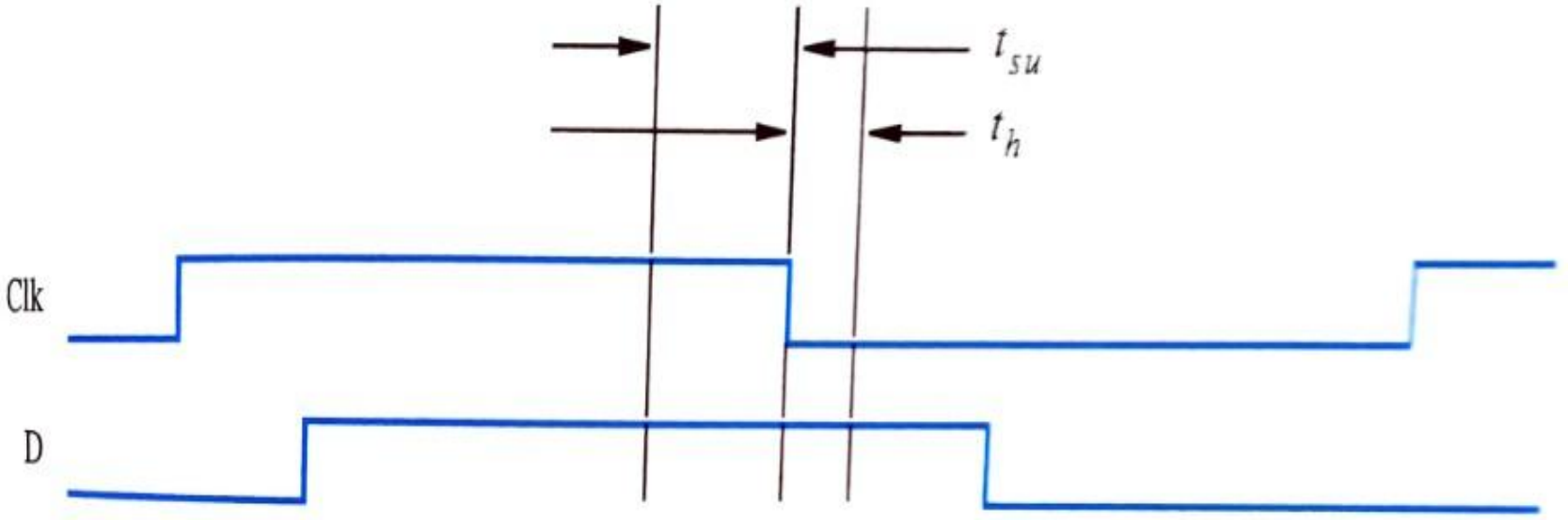
Characteristic Equation

More on the JK flip-flop

- JK flip-flop combines the behaviour of SR flip-flop and a T flip-flop.
- That is, it behaves like the SR flip-flop where J = S and K = R for all the input values except J=K=1.
- Recall that we avoid S=R=1 in SR flip-flop (go back to the diagrams with NOR gates in part 1). Here, however, in the case of J-K flip-flop, it toggles its state like the T flip-flop.
- JK can be used for storage just like D flip-flop. It can also work as T flip-flop when J and K are connected together.

When will our flip-flops operate reliably ?

- Two quantities called setup time (t_{su}) and hold time (t_h) are important for reliable operation.



What exactly are t_{su} and t_h ?

- The setup time (t_{su}) is the minimum time that the D input signal must be stable prior to the clock edge (it is the negative edge in the figure shown).
- The hold time (t_h) is the minimum time that the D input signal must be stable after the clock edge (in the figure, it is the negative edge of the Clk signal).
- We are assuming here that the reference is the negative edge of the Clk signal. We could instead have the positive edge of the signal.

Data input and clock edge …

- The setup and hold time specify how long the data input (D) needs to be stable before and after the clock edge to guarantee that the flip-flop captures the correct data value properly. So we are looking at the input data signal.

Who gives these times ?

- Manufacturers of integrated circuit chips provide the information regarding setup time and hold time on the data sheets that describe their chips.
- For CMOS technology, typical values of setup time and hold time are 3 ns and 2 ns respectively.

How do we exactly use these ?

- We will need to carefully calculate the maximum frequency at which the circuit can operate. Will see examples in tutorial.

Why do we need setup time and hold time ?

- This needs examination at the level of building blocks of a flip-flop.
- In particular, flip-flop building blocks include transmission gates and inverters.
- Need to look at the transistor level structure of a D flip-flop to see how inverters are used to realize a latching circuit.
- One needs to examine the operation of the latching circuit. For now, we will skip the details.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.
- C. Roth and L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021.